Time's Arrow as Non-Invertibility of Record Dynamics

A Mathematical Formalization Connecting Coarse-Graining, Entropy Production, Decoherence, and Temporal Experience

(Explanatory formalization in mathematical language; not axiomatic foundations; not a claim of fundamental time-asymmetry)

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Abstract

We formalize the arrow of time as a structural property of *observer-accessible record dynamics* rather than a statement about fundamental micro-dynamics. Let $(\mathcal{X}, \Sigma, \mu)$ be a fine-grained state space evolving under an invertible, measure-preserving flow Φ_t (classical Hamiltonian dynamics) or unitary evolution (quantum theory). An observer does not access $X_t \in \mathcal{X}$ directly; instead they access records $Y_t = \pi(X_t) \in \mathcal{Y}$ produced by a coarse-graining (projection) $\pi : \mathcal{X} \to \mathcal{Y}$, equivalently by a sub-$\sigma$-algebra $\mathcal{G} \subseteq \Sigma$. Even when Φ_t is reversible, the induced record evolution $F_t = \pi \circ \Phi_t$ is generally non-injective; time reversal becomes a non-identifiable inverse problem on records. Measure-theoretically, coarse-graining converts an invertible group into a forward-time Markov semigroup on the record space. We define an observer-level uncertainty $\mathrm{EE}(t) = H(X_t \mid \mathcal{F}_t^Y)$ where $\mathcal{F}_t^Y = \sigma(Y_s : s \leq t)$ is the record filtration, and show how typical interaction with unobserved degrees of freedom yields effective entropy production and monotone growth of conditional uncertainty (in expectation / with high probability under standard mixing assumptions). We then integrate a discrete *time-anchor* representation of experienced time: anchor events are threshold crossings of a salience functional on records, yielding a multi-scale anchor density, information-geometric structure on anchor-parameter manifolds, and computational bounds for temporal synchronization and reconstruction. The resulting framework provides a single mathematical language linking coarse-grained thermodynamic irreversibility, decoherence-induced classicality, and the phenomenology of temporal flow, while keeping claims explicitly limited to record-level access.

Keywords: arrow of time; coarse-graining; factor maps; Markov semigroups; entropy production; conditional entropy; decoherence; information geometry; time-anchors; computational bounds

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1 Scope, thesis, and claim discipline

1.1 What this paper is (and is not)

This manuscript is written to be legible to mathematically trained readers (ergodic theory, probability, information theory, operator methods) while addressing a physical and phenomenological question: *why does time appear irreversible?* The key discipline is to separate three levels:

- **Fine-grained dynamics (micro-level):** may be reversible (invertible / unitary).
- **Record dynamics (observer-level):** operates on compressed information and is generically non-invertible.
- **Phenomenology (experienced time):** modeled as discrete record-update structure (anchors) rather than a claim about fundamental time being discrete.

This paper does *not*:

- propose a new fundamental time-asymmetric law,
- "prove the Second Law" pointwise for all trajectories,
- claim that information is destroyed at the micro-level.

1.2 Core thesis (structural, not metaphysical)

> **Thesis.** Even when the fine-grained evolution is reversible, the dynamics available to a bounded observer—the evolution of records—is generically non-invertible. This non-invertibility induces an arrow of time in inference, memory, and experience.

The mathematical mechanism is simple and standard in dynamical systems: *factor maps / quotient dynamics / coarse-graining.* The paper's contribution is to (i) present a coherent, journal-ready formal synthesis in one language, (ii) explicitly identify where irreversibility enters (projection), and (iii) connect this to experienced time via a record-based anchor formalism.

2 Fine-grained dynamics

2.1 Classical baseline

Let $(\mathcal{X}, \Sigma, \mu)$ be a probability space representing complete microstates. Let $\Phi_t : \mathcal{X} \to \mathcal{X}$ be a measurable flow. We adopt the standard reversible baseline:

Assumption 2.1 (Invertible, measure-preserving micro-dynamics)**.** *For each $t \in \mathbb{R}$, Φ_t is invertible and measure-preserving:*

$$\Phi_t^{-1} = \Phi_{-t}, \qquad \mu(\Phi_t^{-1}(A)) = \mu(A) \quad \forall A \in \Sigma.$$

This covers Hamiltonian flows preserving Liouville measure in phase space; no arrow is built in.

2.2 Quantum baseline

For quantum theory, the analogous baseline is unitary evolution of a closed system:

$$\rho(t) = U_t \rho(0) U_t^\dagger, \qquad U_t^\dagger U_t = \mathbb{1}.$$

Global evolution is reversible (invertible on density operators by $U_t^\dagger$).

3 Records as coarse-graining: projection, σ-algebras, and quotient dynamics

3.1 Record map and observable σ-algebra

Definition 3.1 (Record map / coarse-graining)**.** *A* record map *is a measurable map $\pi : \mathcal{X} \to \mathcal{Y}$ into a record space $(\mathcal{Y}, \Sigma_\mathcal{Y})$ such that the observer's accessible variables are functions of $Y = \pi(X)$.*

Equivalently, records can be specified by a sub-σ-algebra:

Definition 3.2 (Observable information)**.** *The observer's information at an instant is represented by a sub-σ-algebra $\mathcal{G} \subseteq \Sigma$. A statistic is observable iff it is $\mathcal{G}$-measurable.*

These are equivalent: given π, take $\mathcal{G} = \pi^{-1}(\Sigma_\mathcal{Y})$; given $\mathcal{G}$, one can take π to be the canonical quotient map onto the atoms of $\mathcal{G}$ (up to standard measurability caveats).

3.2 Quotient / factor map construction (what math readers expect)

Define an equivalence relation on $\mathcal{X}$:

$$x \sim x' \quad \Longleftrightarrow \quad \pi(x) = \pi(x').$$

Let $\mathcal{X}/\sim$ be the quotient space of equivalence classes. Then π factors as

$$\mathcal{X} \xrightarrow{q} \mathcal{X}/\sim \xrightarrow{\tilde{\pi}} \mathcal{Y},$$

where q is the canonical quotient map and $\tilde{\pi}$ is injective on the quotient image (or an isomorphism if $\mathcal{Y}$ is chosen as the quotient).

Remark 3.1. *This explicit quotient view matters: "information loss" is literally the collapse of entire equivalence classes into a single record state. The arrow begins at the moment we insist the observer's data live on $\mathcal{Y}$ (or $\mathcal{X}/\sim$) rather than $\mathcal{X}$.*

3.3 Induced record evolution and its non-invertibility

Define the induced record evolution:

$$F_t := \pi \circ \Phi_t.$$

If π is non-injective, F_t is generally not invertible.

Lemma 3.1 (Projection breaks invertibility)**.** *If there exist $x_1 \neq x_2$ with $\pi(x_1) = \pi(x_2)$, then for any t,*

$$F_t(x_1) = F_t(x_2) \quad \textit{whenever} \quad \pi(\Phi_t(x_1)) = \pi(\Phi_t(x_2)).$$

In particular, invertibility on $\mathcal{Y}$ cannot be recovered without extra information beyond records.

This lemma is intentionally modest: it does not claim F_t collapses *every* pair; it asserts the structural fact that record-level invertibility is not guaranteed even if micro-dynamics is invertible.

Proposition 3.1 (Record-level time reversal is an inverse problem)**.** *Fix $t_1 > t_0$ and set $\tau = t_1 - t_0$. The forward observation map is*

$$X_{t_0} \mapsto Y_{t_1} = (\pi \circ \Phi_\tau)(X_{t_0}).$$

If π is non-injective, then the inverse image $(\pi \circ \Phi_\tau)^{-1}(y)$ generically contains multiple candidates; reconstructing X_{t_0} from Y_{t_1} is non-identifiable without additional data.

4 Operator picture: from invertible groups to Markov semigroups

4.1 Koopman / Perron–Frobenius view (fine level)

Define the Koopman operator U_t on observables f by

$$(U_t f)(x) = f(\Phi_t(x)).$$

When Φ_t is measure-preserving, U_t is an isometry on $L^2(\mu)$ and forms a group in t.

4.2 Coarse-graining via conditional expectation

Let $\mathcal{G} \subseteq \Sigma$ be the observable σ-algebra. Coarse-graining an observable f is the $\mathcal{G}$-measurable projection

$$\mathsf{C}f := \mathbb{E}[f \mid \mathcal{G}].$$

This is the canonical probabilistic meaning of "projection" in this paper.

4.3 The induced semigroup on records

Let $Y_t = \pi(X_t)$ be the record process. For a bounded measurable g on $\mathcal{Y}$, define

$$(P_t g)(y) := \mathbb{E}[g(Y_t) \,|\, Y_0 = y]\,.$$

Equivalently, there exists a Markov kernel $K_t(y, dy')$ such that

$$(P_t g)(y) = \int_{\mathcal{Y}} g(y')\, K_t(y, dy').$$

Lemma 4.1 (Group-to-semigroup reduction)**.** *Even if $(\Phi_t)_{t\in\mathbb{R}}$ is an invertible group on $\mathcal{X}$, the induced operators $(P_t)_{t\geq 0}$ on records form (in general) only a* semigroup*:*

$$P_{t+s} = P_t P_s, \qquad t, s \geq 0,$$

and need not admit inverses P_{-t} acting on record observables.

Interpretation. The semigroup property is the mathematical signature of irreversible effective dynamics. Intuitively: losing hidden-state information destroys the ability to "run the movie backward" on the compressed description.

5 Arrow variable: conditional uncertainty under record filtration

5.1 Filtration of accessible information

Define the record filtration

$$\mathcal{F}_t^Y := \sigma(Y_s : s \leq t).$$

This is the information that an idealized observer could have accumulated by time t (assuming perfect storage of records).

5.2 Observer-level uncertainty

Definition 5.1 (Epistemic entropy (observer-level))**.** *Define*

$$\mathrm{EE}(t) := H(X_t \mid \mathcal{F}_t^Y),$$

or, in discrete settings, $\mathrm{EE}(t) = H(X_t \mid Y_{\leq t})$.

Remark 5.1 (Honesty about monotonicity)**.** *We do* not *claim* $\mathrm{EE}(t)$ *increases strictly for every trajectory. The correct claim class in statistical mechanics is* typicality / expectation*: for large systems with mixing hidden degrees of freedom and a fixed coarse-graining,* $\mathrm{EE}(t)$ *tends to be non-decreasing in expectation (or with high probability under a reference measure) until it saturates.*

5.3 Entropy production on record dynamics (relative entropy contraction)

Let ν be a reference distribution on $\mathcal{Y}$ (e.g. the stationary distribution for the Markov kernel K_t when it exists). A standard way to quantify irreversibility is via contraction of relative entropy under Markov operators:

$$D_{\mathrm{KL}}(\rho P_t \,\|\, \nu) \leq D_{\mathrm{KL}}(\rho \,\|\, \nu),$$

which holds for many Markov semigroups under standard conditions (data-processing inequality). This expresses that coarse-grained evolution tends to wash out distinguishability in record space.

Remark 5.2. *This is precisely the "arrow as loss of distinguishability" viewpoint: you can think of time direction as the direction in which statistical distinguishability between candidate pasts collapses under coarse-graining.*

6 System–environment interaction: where records lose information

6.1 Hidden degrees of freedom

Write

$$X_t = (S_t, E_t),$$

system plus environment. Global evolution can be invertible:

$$(S_t, E_t) \mapsto (S_{t+\Delta}, E_{t+\Delta})$$

yet records keep only a function of S_t (and perhaps a small subset of environment channels).

6.2 Mechanism: export of information into unobserved correlations

Interaction typically creates correlations $I(S_t; E_t)$ that are not recorded. The record map π discards these correlations. Therefore, even though micro-information is conserved under invertible dynamics, *record-accessible information* decreases and conditional uncertainty increases.

Proposition 6.1 (Irreversibility from correlation export (conceptual))**.** *When interactions create correlations with unobserved degrees of freedom that are not included in the record filtration $\mathcal{F}_t^Y$, the record-level inverse problem becomes increasingly underdetermined;* $\mathrm{EE}(t)$ *tends to increase (in expectation / typicality) until limited by the record capacity.*

This proposition is intentionally framed as a structural, model-level statement: it is the common core behind thermodynamic coarse-graining and quantum decoherence.

7 Quantum formulation: partial trace as projection

7.1 Closed evolution, open records

Let ρ_{SE} evolve unitarily:

$$\rho'_{SE} = U \rho_{SE} U^\dagger.$$

Records are typically subsystem-restricted (or measurement-restricted):

$$\rho_S = \mathrm{Tr}_E(\rho_{SE}),$$

or more generally POVM outcome distributions.

Lemma 7.1 (Partial trace is many-to-one)**.** *The map $\rho_{SE} \mapsto \rho_S = \mathrm{Tr}_E(\rho_{SE})$ is not injective. Distinct global states can yield identical reduced states, so global micro-information becomes inaccessible at the record level.*

7.2 Decoherence as record-level entropy increase

Subsystem entropy

$$S(\rho_S) = -\mathrm{Tr}(\rho_S \log \rho_S)$$

can increase under unitary global evolution. This is not a contradiction: entropy is transferred into correlations with E, and the record map discards those correlations.

8 Experienced time as discrete record updates: time-anchors

8.1 Anchors as threshold events on records

We model "experienced time" as a sequence of discrete record-update events rather than continuous access to micro-time.

Definition 8.1 (Salience and anchors)**.** *Let $s : \mathcal{Y} \to \mathbb{R}_{\geq 0}$ be a salience functional and $\theta > 0$ a threshold. An* anchor event *occurs at times t_i such that*

$$s(Y_{t_i}) > \theta.$$

The experienced timeline is the ordered anchor sequence $\{t_i\}$.

Remark 8.1. *This is a modeling layer: it asserts that the observer's effective time-indexing is organized by discrete record updates. It does* not *assert that physical time is discrete.*

8.2 Anchor density and relativistic reparameterizations (model equations)

Let $\lambda(t)$ be anchor density (rate of anchor formation). A simple model relating anchor rate and relativistic factors uses the Lorentz factor $\gamma = (1 - v^2/c^2)^{-1/2}$:

$$\theta^{\text{rel}}_{\text{salience}} = \theta^{\text{rest}}_{\text{salience}} \cdot \gamma, \tag{A1}$$

$$\lambda_{\text{rel}}(v) = \lambda_{\text{rest}} \cdot \gamma. \tag{A2}$$

In curved spacetime with metric component g_{00}:

$$d\tau = \sqrt{-g_{00}}\, dt, \tag{A3}$$

$$\theta_{\text{salience}}(r) = \theta^{\infty}_{\text{salience}} \sqrt{-g_{00}(r)}. \tag{A4}$$

These express a model-level coupling between record thresholds and the time parameterization used in physics.

8.3 Multi-scale anchor density and cross-scale coupling

Multi-scale anchoring:

$$\lambda_{\text{total}}(t) = \sum_{k=1}^{N} w_k\, \lambda_k(t_k). \tag{A5}$$

Cross-scale coupling:

$$\frac{\partial \lambda_k}{\partial t} = f_k(\lambda_k) + \sum_{j \neq k} C_{kj} \big(\lambda_j - \lambda_k\big). \tag{A6}$$

Nested-interval consistency (one useful constraint):

$$\int_{\tau_k} \lambda_k(t)\, dt = \int_{\tau_{k+1}} \lambda_{k+1}(t)\, dt, \qquad \tau_k \subset \tau_{k+1}. \tag{A7}$$

9 Quantum anchoring and salience: operator-level modeling

9.1 Quantum time-anchor operator (model)

Define a quantum anchor operator

$$\hat{A}(t) = \sum_i |\psi_i\rangle \langle\psi_i| \otimes |t_i\rangle \langle t_i| \,, \tag{Q1}$$

where $|\psi_i\rangle$ are internal states and $|t_i\rangle$ represent anchor labels.

9.2 Salience via quantum Fisher information (model)

Model salience thresholding via quantum Fisher information:

$$\theta_{\text{salience}} = F_Q\left[\rho(t), \frac{\partial \rho}{\partial t}\right]. \tag{Q2}$$

Define a quantum anchor density

$$\lambda_{\text{quantum}} = \mathrm{Tr}[\Pi_{\text{salience}} \rho(t)] \,, \tag{Q3}$$

where Π_{salience} projects onto a high-salience subspace.

Remark 9.1. *Equations (Q1)–(Q3) are modeling commitments: they propose a concrete bridge from quantum distinguishability to record-update events. They should be treated as hypotheses to be calibrated empirically.*

10 Information geometry on anchor space

Let $p(t \mid \lambda)$ be the anchor-formation probability conditioned on parameters $\lambda = (\lambda^1, \ldots, \lambda^m)$. Define Fisher information metric:

$$g_{ij}(\lambda) = \mathbb{E}\left[\frac{\partial \log p(t \mid \lambda)}{\partial \lambda_i} \frac{\partial \log p(t \mid \lambda)}{\partial \lambda_j}\right]. \tag{G1}$$

Geodesic dynamics on parameter manifold:

$$\frac{d^2 \lambda^i}{d\tau^2} + \Gamma^i{}_{jk} \frac{d\lambda^j}{d\tau} \frac{d\lambda^k}{d\tau} = 0. \tag{G2}$$

This provides a geometric language for trajectories of record-density parameters, potentially useful for describing "optimal" adaptations of the observer's anchoring process.

11 Computational irreversibility: bounds for reconstruction and synchronization

Even when micro-information is conserved, record-level reversal can fail for two reasons:

- **Information-theoretic:** the record map is non-injective (non-identifiability).
- **Computational:** even approximate inversion can be intractable.

11.1 Cost-based reconstruction model (generic)

Consider a cost functional over anchor sequences:

$$C = \sum_i \mathrm{cost}(t_i, t_{i+1}) + \lambda_{\mathrm{penalty}}(\lambda_{\mathrm{actual}} - \lambda_{\mathrm{target}}). \tag{C1}$$

For general temporally dependent costs, optimizing anchor sequences is computationally hard (a standard phenomenon in combinatorial optimization on sequences).

11.2 Resource scaling statements (model bounds)

Over a temporal window T with anchor rate λ:

$$\text{space } O(\lambda T \log T), \qquad \text{time } O(\lambda^2 T).$$

For n agents synchronizing anchor processes, a representative communication lower bound form is:

$$\Omega(n\lambda \log \lambda) \text{ bits.} \tag{C2}$$

These bounds are included to emphasize that "irreversibility" can be enforced in practice even if micro-information is not literally destroyed.

12 Synthesis: the arrow as record non-invertibility (final statement)

We can now state the paper's central content precisely and honestly.

Proposition 12.1 (Arrow-of-time as record non-invertibility (structural statement))**.** *Assume reversible micro-dynamics Φ_t on $(\mathcal{X}, \Sigma, \mu)$ and a non-injective record map $\pi : \mathcal{X} \to \mathcal{Y}$ (or observable σ-algebra $\mathcal{G}$). Then:*

1. *The induced record evolution $F_t = \pi \circ \Phi_t$ is generally non-invertible; time reversal on records is a non-identifiable inverse problem.*

2. *The induced record process admits a Markov semigroup $(P_t)_{t\geq 0}$ on record observables, reflecting effective irreversibility at the observer level.*
3. *Under repeated interaction with unobserved degrees of freedom and standard mixing/typicality conditions, conditional uncertainty*

$$\mathrm{EE}(t) = H(X_t \mid \mathcal{F}_t^Y)$$

is typically non-decreasing (in expectation / with high probability) until saturation, providing an information-theoretic arrow variable.

Corollary 12.1 (Interpretation)**.** *The "direction of time" for the observer is the direction in which accessible distinguishability collapses and conditional uncertainty accumulates under record-level evolution.*

13 Empirical and conceptual implications (what would count as support)

13.1 What this framework predicts qualitatively

- **Decoherence and time perception:** if anchor salience is tied to quantum (or classical) distinguishability measures, regimes of faster decoherence / faster record-update should correlate with faster experienced temporal progression (model-dependent).
- **Multi-scale coupling:** cross-scale coupling (A6) predicts structured correlations between fast and slow anchoring processes (e.g. attention-level vs memory-level timescales).
- **Compression dependence:** changing the coarse-graining (sensor resolution, memory capacity, summarization) changes the arrow variable $\mathrm{EE}(t)$ and the anchor density $\lambda(t)$.

13.2 What would falsify key modeling layers

- If anchor events cannot be described as threshold crossings of any stable salience functional on records, the anchor model is wrong (even if record non-invertibility remains correct).
- If multi-scale anchor densities do not exhibit consistent coupling signatures, the multi-scale hypothesis is unsupported.
- If record-level irreversibility were absent under clear projection (non-injective π) and hidden variables, that would contradict a standard body of results in coarse-grained dynamics; such findings would require revisiting assumptions about observability and hidden-state modeling.

14 Limitations (explicit)

- **Typicality dependence:** monotonic growth of $\mathrm{EE}(t)$ is a typicality/expectation claim under mixing assumptions, not a pointwise theorem.
- **Model dependence of anchors:** equations (A1)–(A7), (Q1)–(Q3), (G1)–(G2), (C1)–(C2) are modeling choices; they should be treated as hypotheses requiring empirical calibration.
- **Choice of record map:** π (or $\mathcal{G}$) is not unique; different observers or instruments imply different record dynamics and thus different arrows.

15 Conclusion

This paper provides a single, mathematically explicit explanation of the arrow of time: *irreversibility arises when reversible micro-dynamics is viewed through a non-injective projection into observer-accessible records.* In that setting, time reversal is not blocked by physics but by *non-identifiability* and *semigroup* evolution on record space. Thermodynamic entropy increase, decoherence, and experienced temporal flow can be expressed in one language: conditional uncertainty under a growing record filtration, Markov semigroups induced by hidden degrees of freedom, and discrete anchor updates organizing experience.

Author note. This manuscript is intended as a journal-style synthesis written in mathematical language. It does not claim axiomatic foundations for time nor a new fundamental time-asymmetric law.

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